LogicSetTheoryReview1

Zadanie	1	2	3	4	5	6	7	8	9	10	11	12	13	14	
Punkty															
Zadanie	**15**	**16**	**17**	**18**	**19**	**20**									**Suma**
Punkty															

Zadanie 1:

1.3 For how many assignments the formula is true?

i) $[(p \Rightarrow q) \Rightarrow \sim (p \Rightarrow r)] \Rightarrow [\sim (p \Rightarrow r) \Rightarrow (p \Rightarrow q)]$

l) $(p \wedge q \wedge \sim r) \Leftrightarrow [(\sim p \Rightarrow \sim q) \vee r]$

o) $(p \Rightarrow q) \Rightarrow [(q \Rightarrow r) \Rightarrow (p \Rightarrow r)]$.

Zadanie 2:

1.6 The same claim with variables varied through set $\mathbb{R}$:

d) *each linear equation has a solution* $(=, 0, \cdot, +)$,

Zadanie 3:

Define the connectives: negation, disjunction, conjunction and implication in terms of

a) Sheffer's functor $|$, where $p \mid q \Leftrightarrow \sim p \vee \sim q$

Zadanie 4:

1.2 Let $x, y, z, t \in \{\mathbf{0}, \mathbf{1}\}$. Solve the following equations:

b) $\sim (x \Rightarrow y) \wedge (x \vee y) = \mathbf{1}$

c) $\sim [(x \vee y \vee t) \Rightarrow (\sim x \vee z \vee \sim t)] = \mathbf{0}$

Zadanie 5:

1.5 Write the mathematical formulas corresponding to the following statements with the help of the following signs only: propositional connectives, quantifiers, variables varied through set $\mathbb{N}$ and symbols indicated in brackets (define auxiliary symbols if necessary)

i) *there doesn't exist the greatest number* $(\leq)$,

Zadanie 6:

1.3 For how many assignments the formula is true?

k) $\sim [(p \vee q \vee r) \Rightarrow (\sim p \wedge \sim q \wedge \sim r)]$

n) $[(p \Rightarrow q) \Rightarrow r] \vee [(\sim r \vee q) \wedge r]$,

Zadanie 7:

Define the connectives: negation, disjunction, conjunction and implication in terms of

a) Sheffer's functor $|$, where $p \mid q \Leftrightarrow \sim p \vee \sim q$

Zadanie 8:

1.5 Write the mathematical formulas corresponding to the following statements with the help of the following signs only: propositional connectives, quantifiers, variables varied through set $\mathbb{N}$ and symbols indicated in brackets (define auxiliary symbols if necessary)

j) *every even number is a sum of two squares*, $(=, \cdot, +)$,

n) *there exists a number which is not a square of an odd number* $(=, \cdot, +)$,

Zadanie 9:

1.3 For how many assignments the formula is true?

j) $[(p \Rightarrow \sim q) \Rightarrow \sim p] \Rightarrow [p \Rightarrow (q \Rightarrow p)]$

Zadanie 10:

1.6 The same claim with variables varied through set $\mathbb{R}$:

b) *the product of two numbers with different signs is a positive number* $(<, \cdot, 0)$,

Zadanie 11:

1.8 Interpret symbols $\alpha, *$ occurring in following formulas to make them (1) true, (2) false.

b) $(\forall x, y, z)[\alpha(x, y) \wedge \alpha(y, z) \Rightarrow \alpha(x, z)]$

Zadanie 12:

1.3 For how many assignments the formula is true?

a) $[p \vee (q \Rightarrow r)] \wedge [\sim r \vee (q \Rightarrow p)]$

Zadanie 13:

1.1 Calculate, according to truth - tables:

e) $(\mathbf{1} \Rightarrow \mathbf{0}) \Leftrightarrow [(\mathbf{0} \Rightarrow \mathbf{1}) \vee \mathbf{0}]$.

Zadanie 14:

1.5 Write the mathematical formulas corresponding to the following statements with the help of the following signs only: propositional connectives, quantifiers, variables varied through set $\mathbb{N}$ and symbols indicated in brackets (define auxiliary symbols if necessary)

b) *x is even* $(=, +)$,

m) *there are no numbers with at most two multiples being squares* $(=, <, \cdot)$,

Zadanie 15:

1.5 Write the mathematical formulas corresponding to the following statements with the help of the following signs only: propositional connectives, quantifiers, variables varied through set $\mathbb{N}$ and symbols indicated in brackets (define auxiliary symbols if necessary)

d) *x is divisible by 3* $(=, +)$,

r) *x is greater than y* $(=, +, \cdot)$

Zadanie 16:

1.7 Prove the following rules for formulas with bounded quantifiers (derive them from suitable logical rules for unbounded quatifiers):

a) $[(\forall x)_{\alpha(x)}\phi(x) \wedge (\forall x)_{\alpha(x)}\psi(x)] \Leftrightarrow (\forall x)_{\alpha(x)}(\phi(x) \wedge \psi(x))$

Zadanie 17:

1.3 For how many assignments the formula is true?

a) $[p \vee (q \Rightarrow r)] \wedge [\sim r \vee (q \Rightarrow p)]$

c) $p \Rightarrow [\sim p \Rightarrow (p \Rightarrow \sim p)]$

Zadanie 18:

1.3 For how many assignments the formula is true?

b) $\sim [p \wedge (r \Rightarrow q)] \vee [q \Rightarrow (p \wedge r)]$

e) $(p \vee \sim q) \wedge (\sim p \vee q) \wedge (p \vee q) \wedge (\sim p \vee \sim q)$

g) $[(p \Rightarrow q) \Rightarrow r] \Rightarrow [(p \Rightarrow r) \Rightarrow (q \Rightarrow r)]$

Zadanie 19:

1.6 The same claim with variables varied through set $\mathbb{R}$:

g) *each system of two linear equations with two variables has a unique solution* $(=, 0, \cdot, +)$,

Zadanie 20:

1.8 Interpret symbols $\alpha, *$ occurring in following formulas to make them (1) true, (2) false.

c) $(\forall x, y)\alpha(x, y) \Leftrightarrow (\forall x)\alpha(x, x)$